PAPER 2

PART – I : CHEMISTRY

M2

SECTION – I (Single Correct Choice Type)

1. The complex showing a spin-only magnetic moment of 2.82 B.M. is

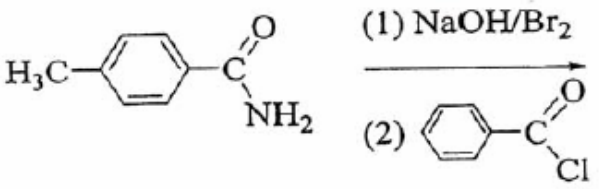
   A) $Ni(CO)_4$ B) $[NiCl_4]^{2-}$ C) $Ni(PPh_3)_4$ D) $[Ni(CN)_4]^{2-}$

ANSWER: B

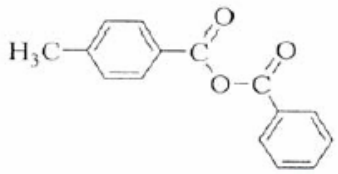
2. The species having pyramidal shape is

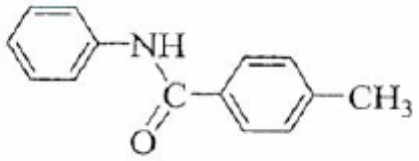
   A) SO_3 B) BrF_3 C) SiO_3^{2-} D) OSF_2

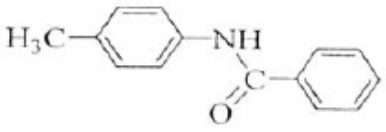
ANSWER: D

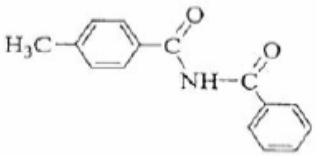
3. In the reaction 

the structure of the Product T is

A) 

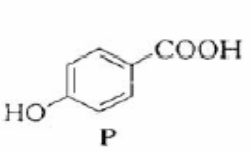
B) 

C) 

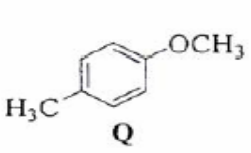
D) 

ANSWER: C

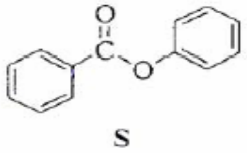
4. The compounds P, Q and S



P

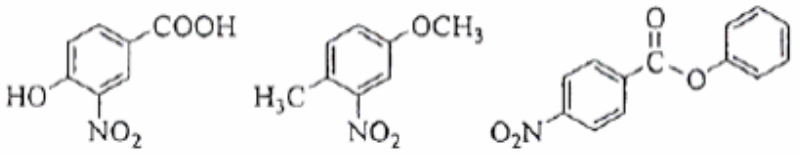


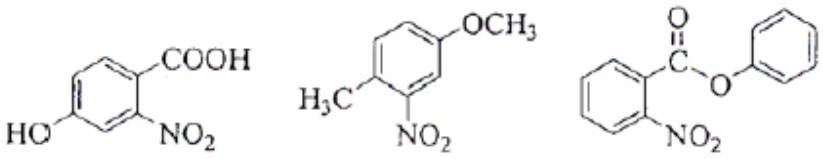
Q

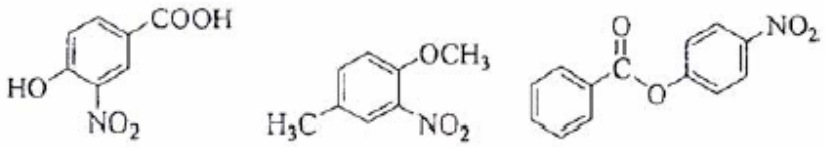


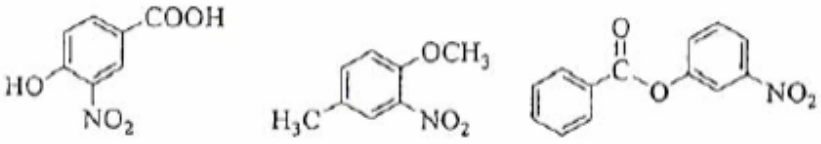
S

were separately subjected to nitration using HNO_3/H_2SO_4 mixture. The major product formed in each case respectively, is

A) 

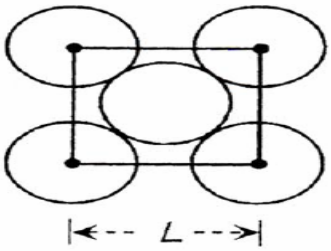
B) 

C) 

D) 

ANSWER: C

5. The packing efficiency of the two-dimensional square unit cell shown below is



A) 39.27% B) 68.02% C) 74.05% D) 78.54%

ANSWER: D

6. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is

A) 1 and diamagnetic
C) 1 and paramagnetic
B) 0 and diamagnetic
D) 0 and paramagnetic

ANSWER: A

SECTION – II (Integer Type)

7. The total number of diprotic acids among the following is

H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$
H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3

ANSWER: 6

8. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is

ANSWER: 3

9. Among the following, the number of elements showing only one non-zero oxidation state is

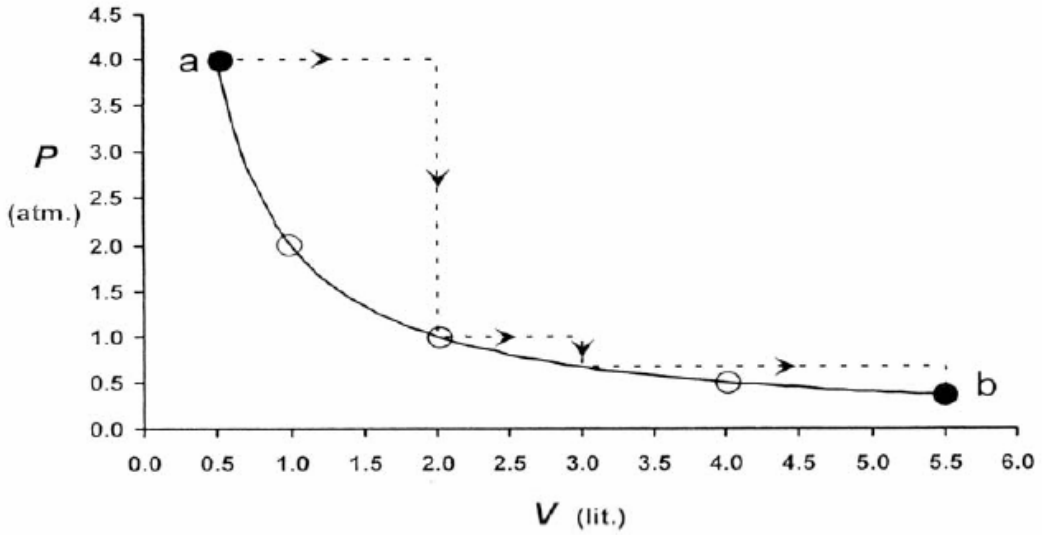
O, Cl, F, N, P, Sn, Tl, Na, Ti

ANSWER: 2

10. Silver (atomic weight = 108 g mol^{-1}) has a density of 10.5 g cm^{-3}. The number of silver atoms on a surface of area 10^{-12} m^2 can be expressed in scientific notation as $y \times 10^x$. The value of x is

ANSWER: 7

11. One mole of an ideal gas is taken from **a** to **b** along two paths denoted by the solid and the dashed lines as shown in the graph below. If the work done along the solid line path is w_s and that along the dotted line path is w_d, then the integer closest to the ratio w_d/w_s is

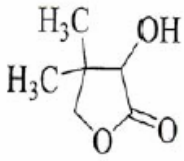


ANSWER: 2

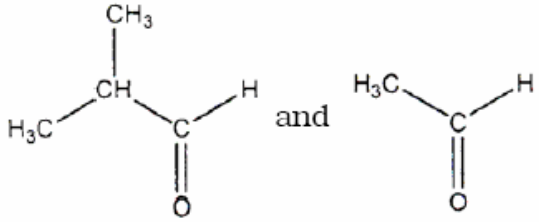
SECTION – III (Paragraph Type)

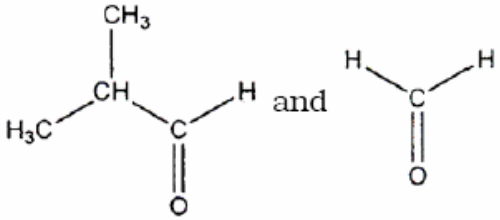
Paragraph for Questions 12 to 14.

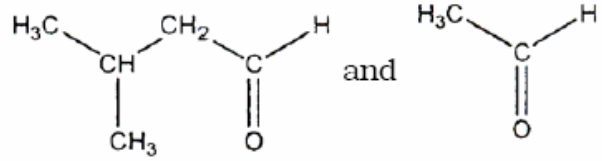
Two aliphatic aldehydes P and Q react in the presence of aqueous K_2CO_3 to give compound R, which upon treatment with HCN provides compound S. On acidification and heating, S gives the product shown below :

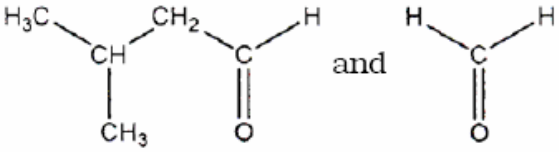


12. The compounds P and Q respectively are

A) 

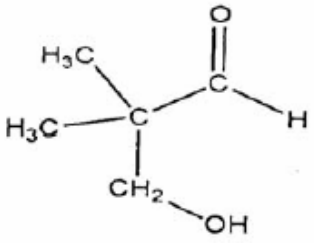
B) 

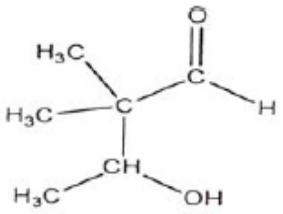
C) 

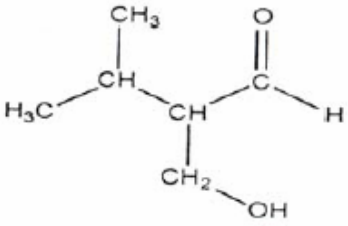
D) 

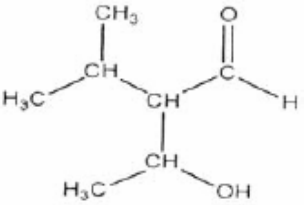
ANSWER: B

13. The compound R is

A) 

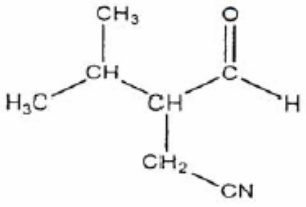
B) 

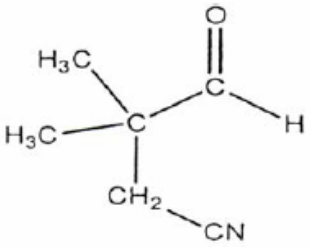
C) 

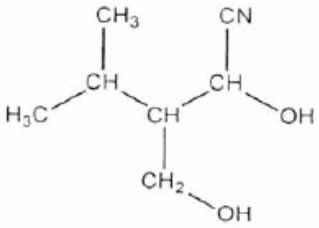
D) 

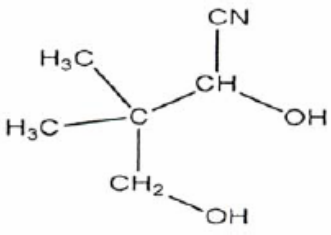
ANSWER: A

14. The compound S is

A) 

B) 

C) 

D) 

ANSWER: D

Paragraph for Questions 15 to 17.

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

15. The state S_1 is

A) 1s B) 2s C) 2p D) 3s

ANSWER: B

16. Energy of the state S_1 in units of the hydrogen atom ground state energy is

A) 0.75 B) 1.50 C) 2.25 D) 4.50

ANSWER: C

17. The orbital angular momentum quantum number of the state S_2 is

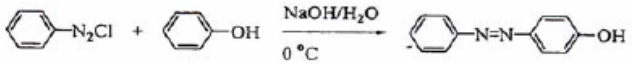
A) 0 B) 1 C) 2 D) 3

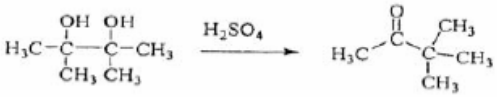
ANSWER: B

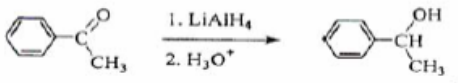
SECTION – IV (Matrix Type)

18. Match the reactions in Column I with appropriate options in Column II.

Column I

A) 

B) 

C) 

D) 

Column II

p) Racemic mixture

q) Addition reaction

r) Substitution reaction

s) Coupling reaction

t) Carbocation intermediate

ANSWER:

A: r and s
B: t
C: p and q
D: r

19. All the compounds listed in **Column I** react with water. Match the result of the respective reactions with the appropriate options listed in **Column II**.

Column I	**Column II**
A) $(CH_3)_2SiCl_2$	p) Hydrogen halide formation
B) XeF_4	q) Redox reaction
C) Cl_2	r) Reacts with glass
D) VCl_5	s) Polymerization
	t) O_2 formation

ANSWER: **A: p and s**
B: p and q and r and t
C: p and q
D: p

PART – II : MATHEMATICS

SECTION – I (Single Correct Choice Type)

20. For r = 0, 1, ..., 10, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then

$$\sum_{r=1}^{10} A_r \left(B_{10}B_r - C_{10}A_r\right)$$

is equal to

A) $B_{10} - C_{10}$ B) $A_{10}\left(B_{10}^2 - C_{10}A_{10}\right)$ C) 0 D) $C_{10} - B_{10}$

ANSWER: D

21. Let S = {1, 2, 3, 4}. The total number of unordered pairs of disjoint subsets of S is equal to

A) 25 B) 34 C) 42 D) 41

ANSWER: D